# Neanderthals

400,000 - 40,000 years ago

Photo: Photaro, World History Encyclopedia (CC BY-SA 4.0)

# Neanderthals

A few thousand breeding adults

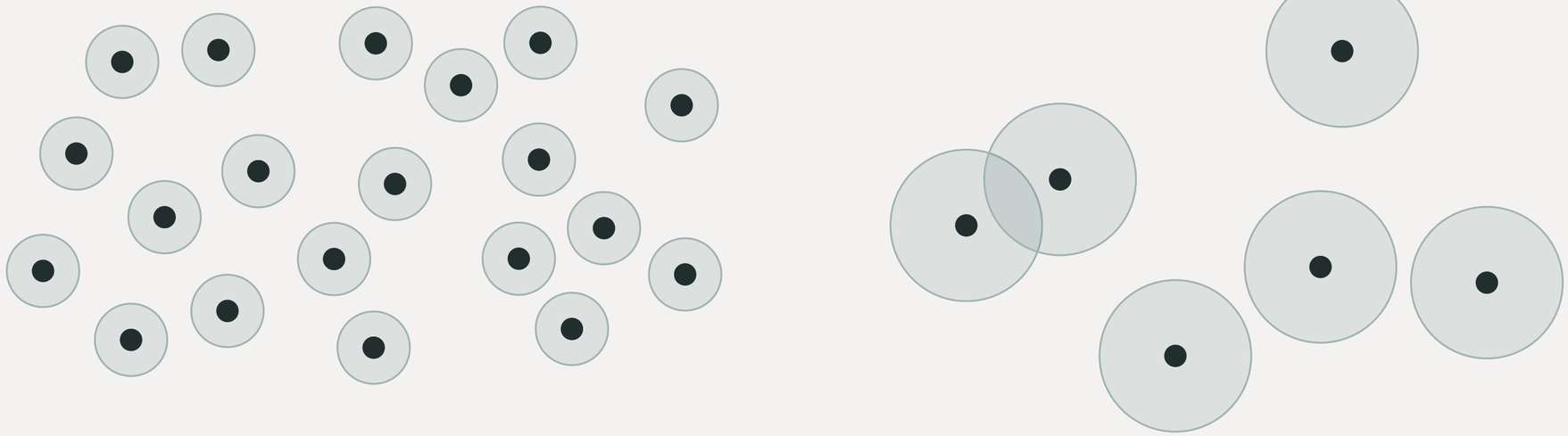
one small band

Band positions illustrative · Vaesen et al. 2019 · Drawn with AI assistance (Claude)

Many neighbours

Few neighbours

## Competitors
more for each



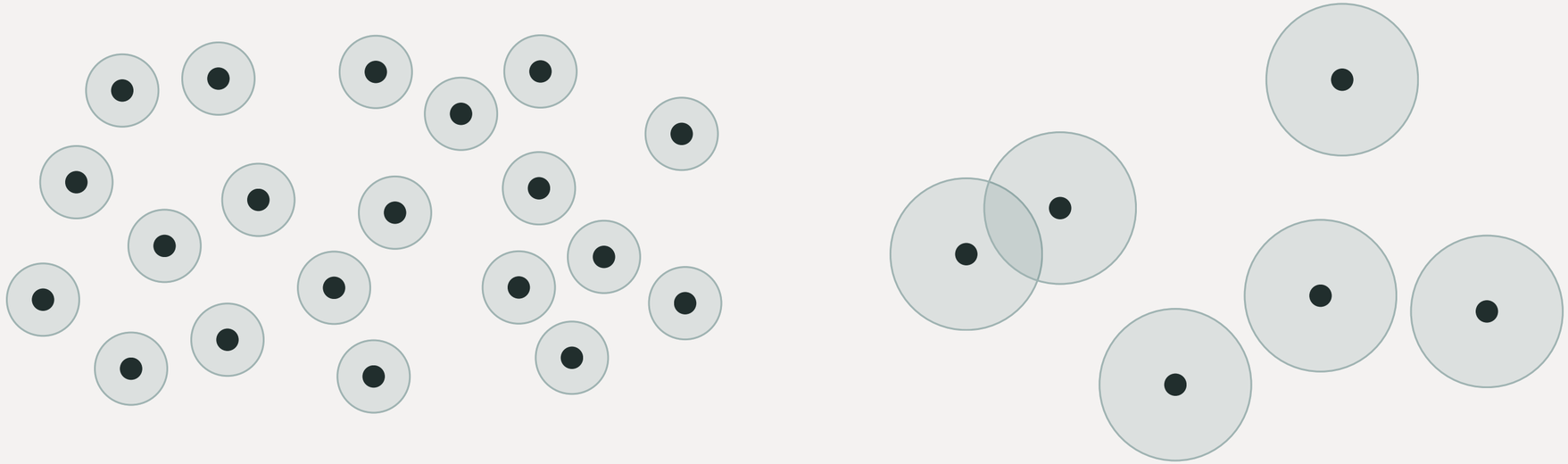
● one individual ○ its food and space

Schematic after Stephens et al. 1999

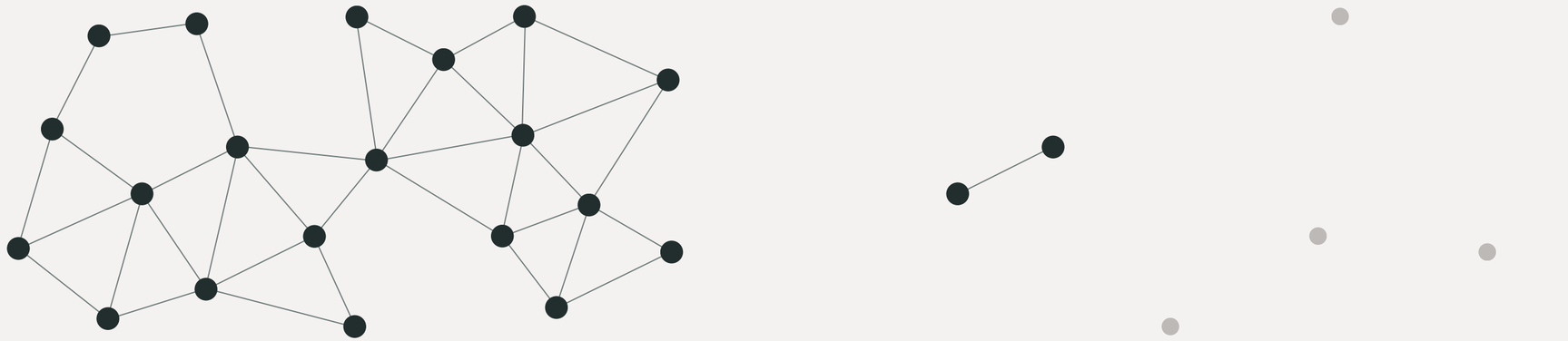
Many neighbours

Few neighbours

## Competitors
more for each



## Partners
fewer helpers



● one individual ○ its food and space — a partner

Schematic after Stephens et al. 1999

Neanderthals

In the simulation

one small band

Band positions illustrative · Vaesen et al. 2019 · Drawn with AI assistance (Claude)

Neanderthals

In the simulation

one small band

Band positions illustrative · Vaesen et al. 2019 · Drawn with AI assistance (Claude)

The smaller the population, the lower each individual's chance of surviving or reproducing.

Source: Stephens et al. 1999

# Mechanisms of the Allee effect

## Mates

Sea urchins

## Safety

Island marmots

## Teamwork

African wild dogs

## Changing habitat

Bark beetles

Sources: Gascoigne et al. 2009 · Brashares et al. 2010 · Courchamp et al. 1999 · Branco et al. 2024

Photos: see image credits

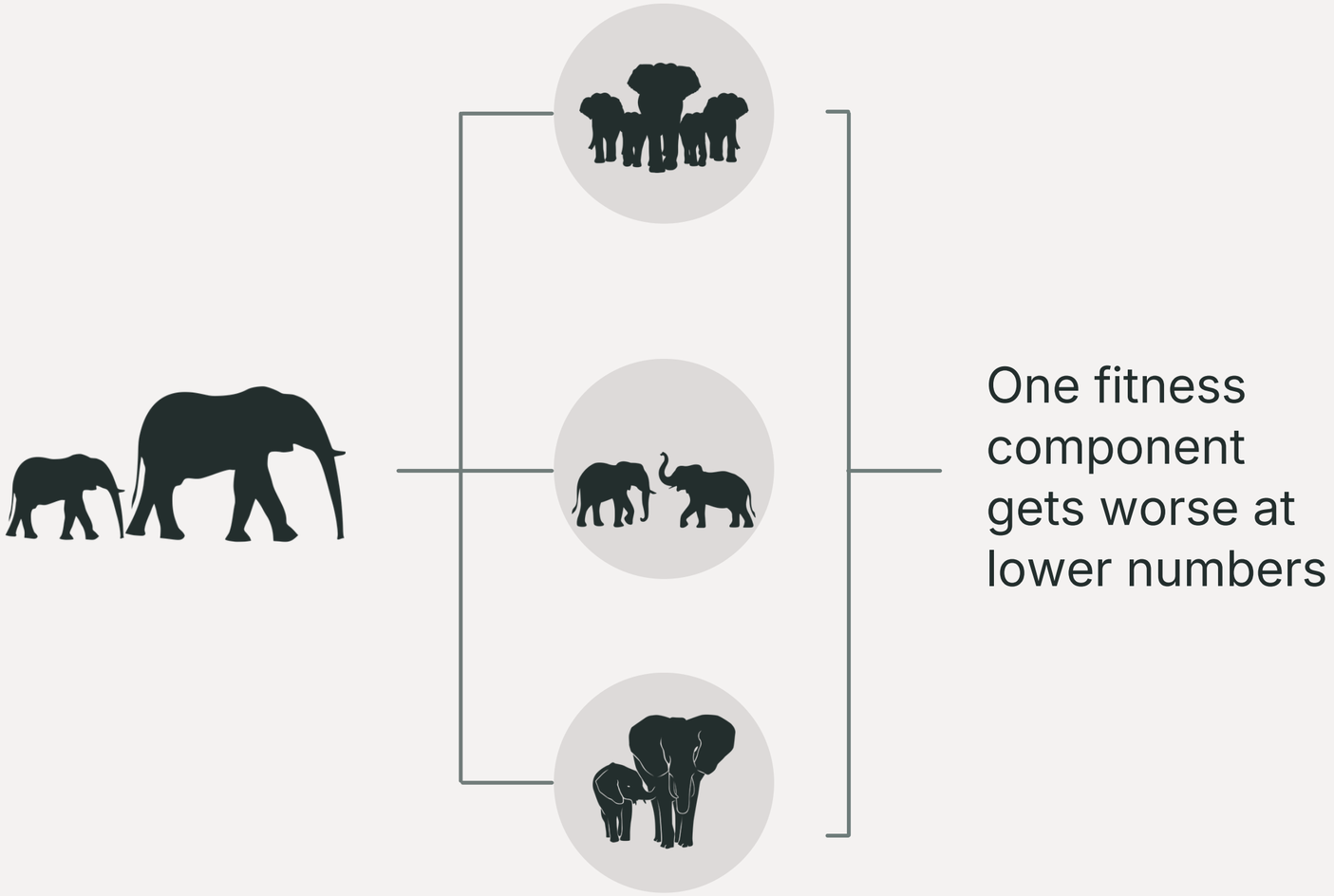
# Are Allee effects an ecological curiosity, or a fundamental driver of population dynamics?

# Component Allee effect

> “A positive relationship between a specific component of individual fitness and population density”
>
> – Muir et al. 2024

Examples of fitness components

Reproductive success
Survival
Mate finding
Offspring survival

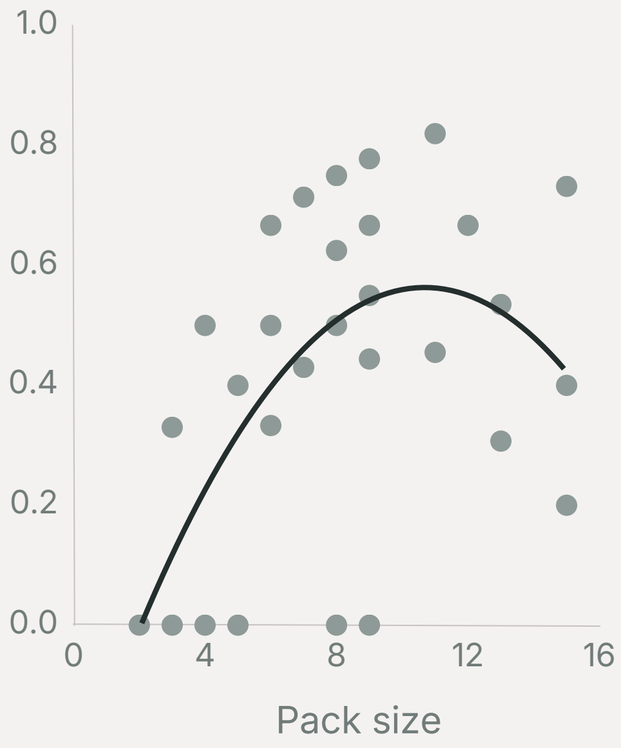



Quote: Muir et al. 2024, p. 2, after Stephens et al. 1999

# Component Allee effect: African wild dog

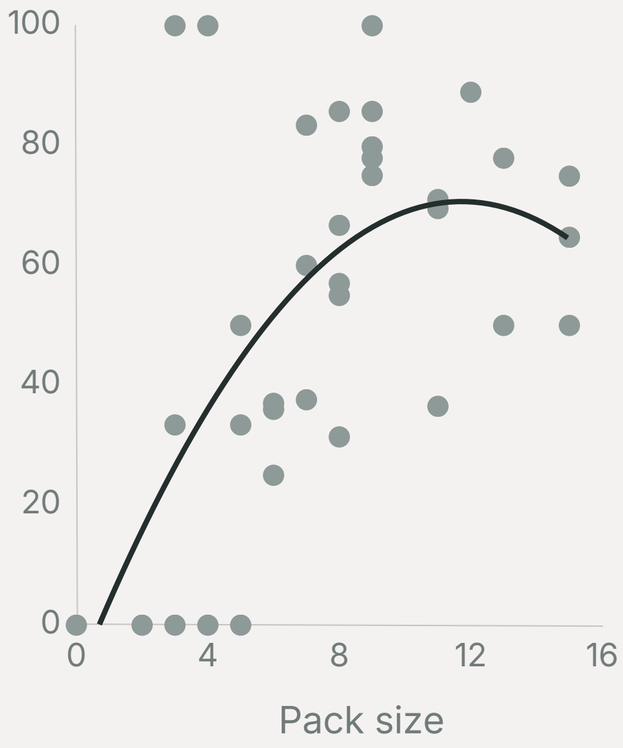
## Pups produced per dog

per capita productivity




## Pup survival

% of pups surviving




Each dot: one pack in one year

Photo: Sabi Sabi Private Game Reserve, via Africa Endeavours

### Why pack size matters

Hunt cooperatively, defend each other, help care for pups

### Fewer pack members

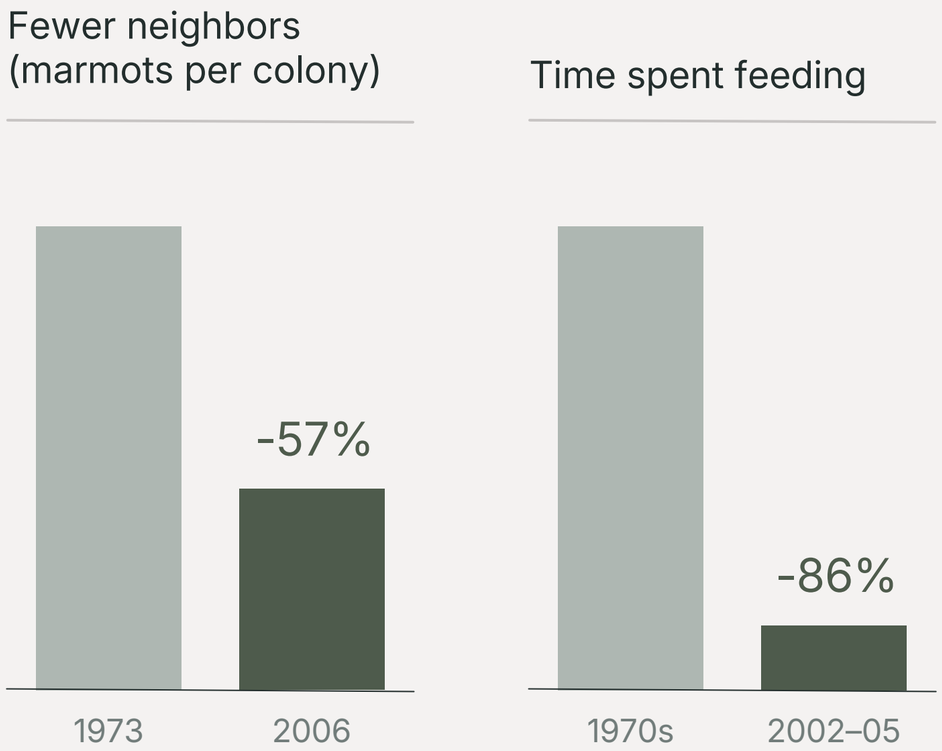
Less cooperative support, so lower reproductive success and pup survival

Data: Angulo et al. 2013, Fig. 2B, 2F; values digitised, approximate. Drawn with AI assistance (Claude)

# Component and demographic Allee effects: Vancouver Island marmot

## Component Allee effect

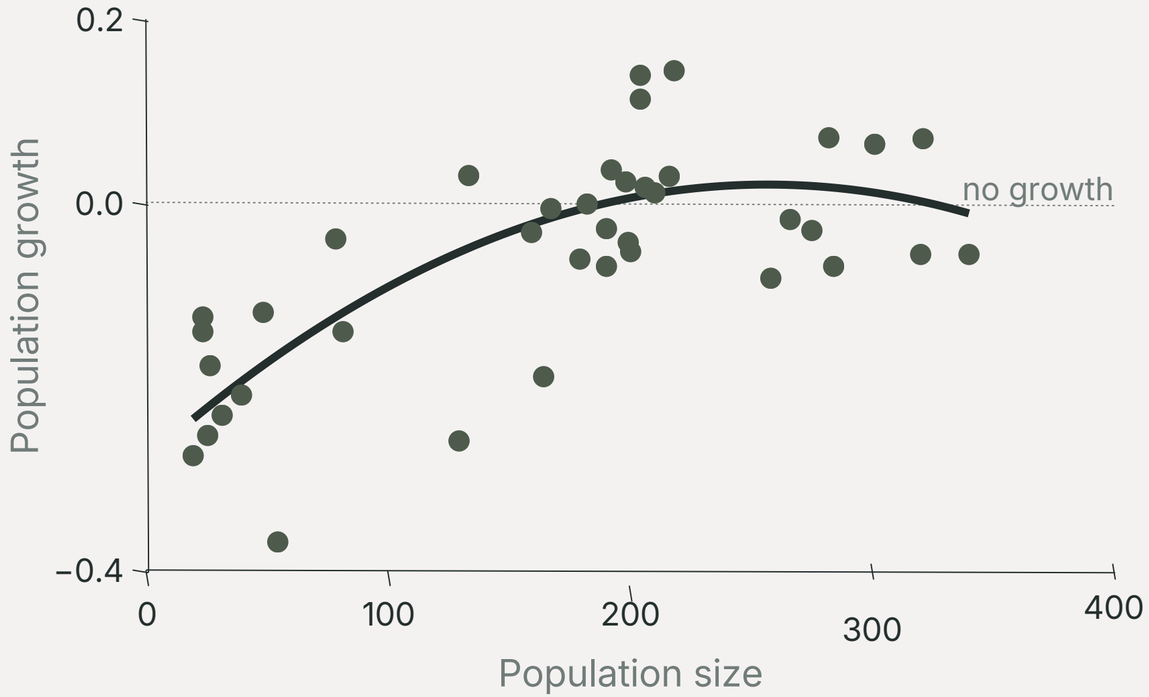
Each marmot does worse




–57%: our calculation, 8.3 to 3.6 marmots per colony

## Demographic Allee effect

The whole population grows more slowly

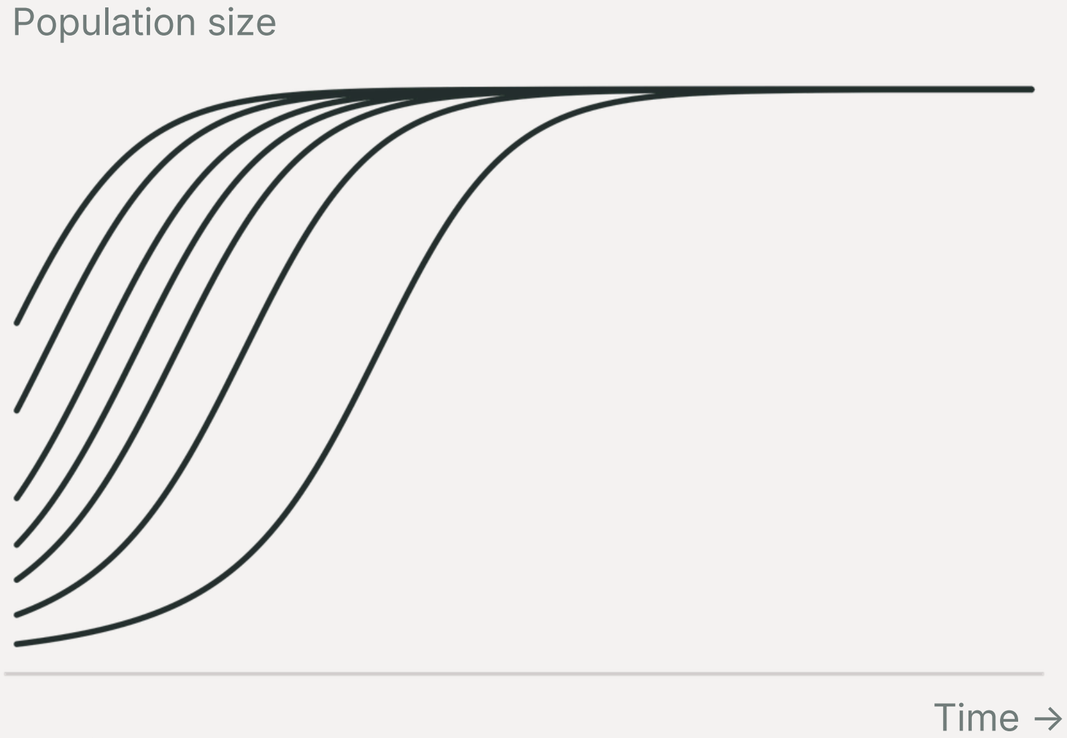



Each dot: one year, 1970–2007. Curve: quadratic fit, $R^2 = 0.55$

Data: Brashares et al. 2010, Fig. 1 and p. 966; terms: Stephens et al. 1999. Drawn with AI assistance (Claude)

# Strong and weak Allee effects

Weak: small populations grow more slowly




Each line: one population

## Flour beetle

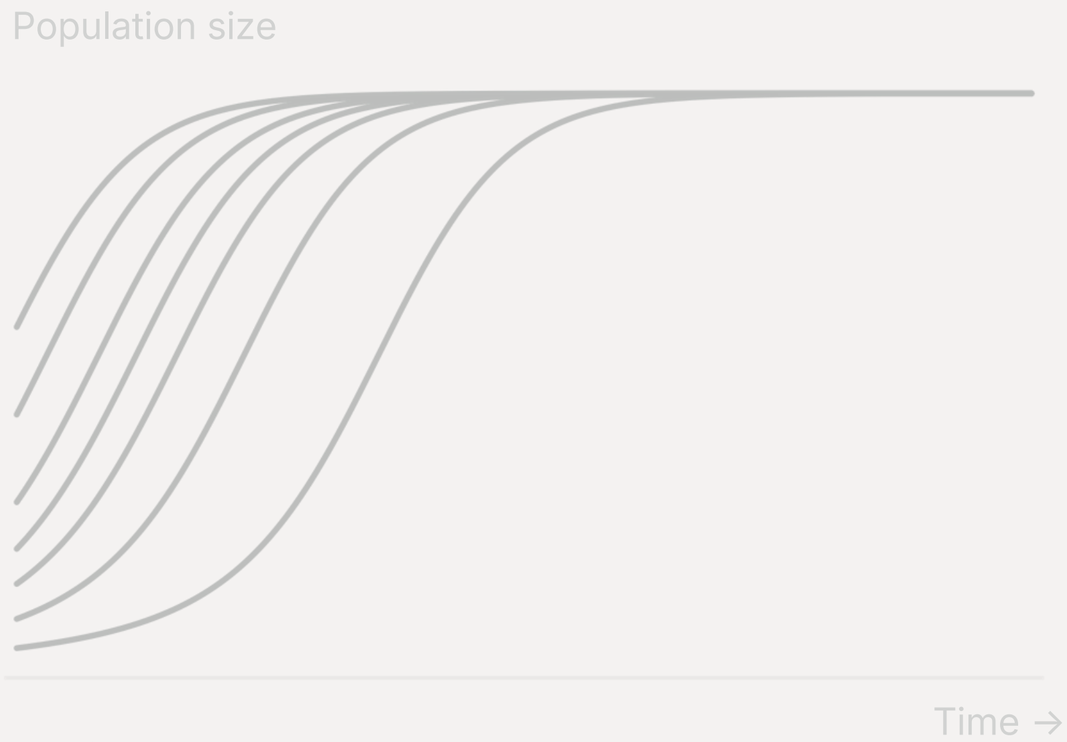
Offspring per female rise with density: a component effect that can add up to a weak Allee effect.

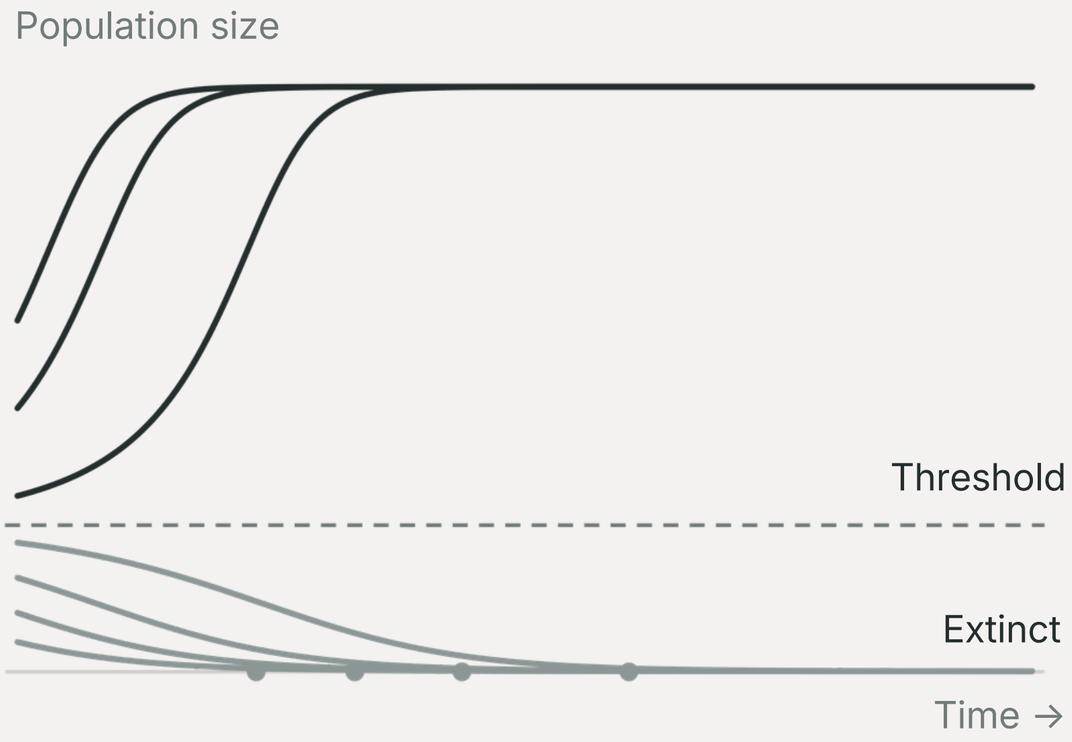
Schematic after Drake & Kramer 2011; flour beetles: Allee 1938, in Courchamp et al. 1999. Drawn with AI assistance (Claude)

# Strong and weak Allee effects

Weak: small populations grow more slowly

Strong: below threshold, more die than are born

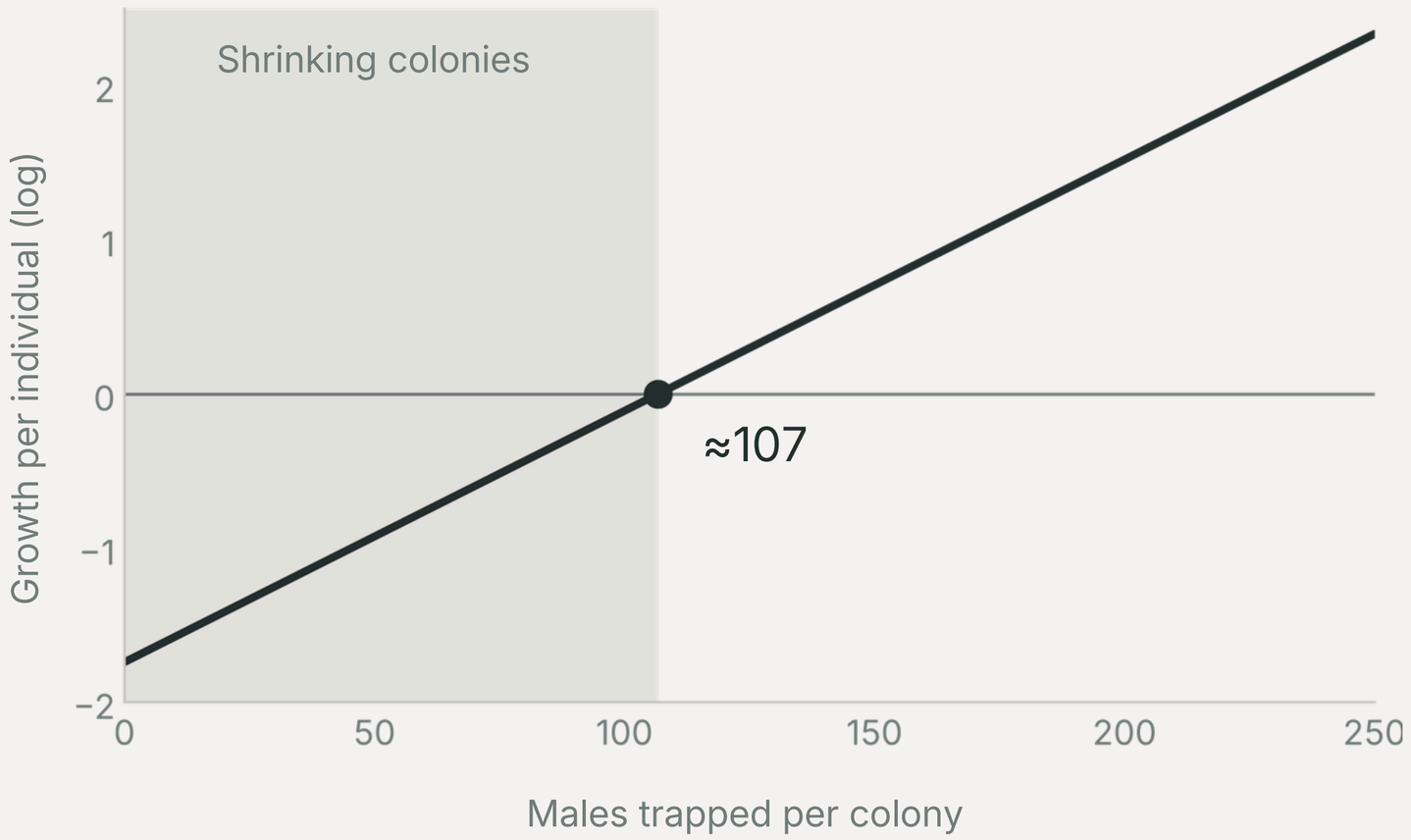



Each line: one population

Schematic after Drake & Kramer 2011; flour beetles: Allee 1938, in Courchamp et al. 1999. Drawn with AI assistance (Claude)

# Strong and weak Allee effects: a real threshold




225 colony-years of trap data from new colonies, USA; data points not shown

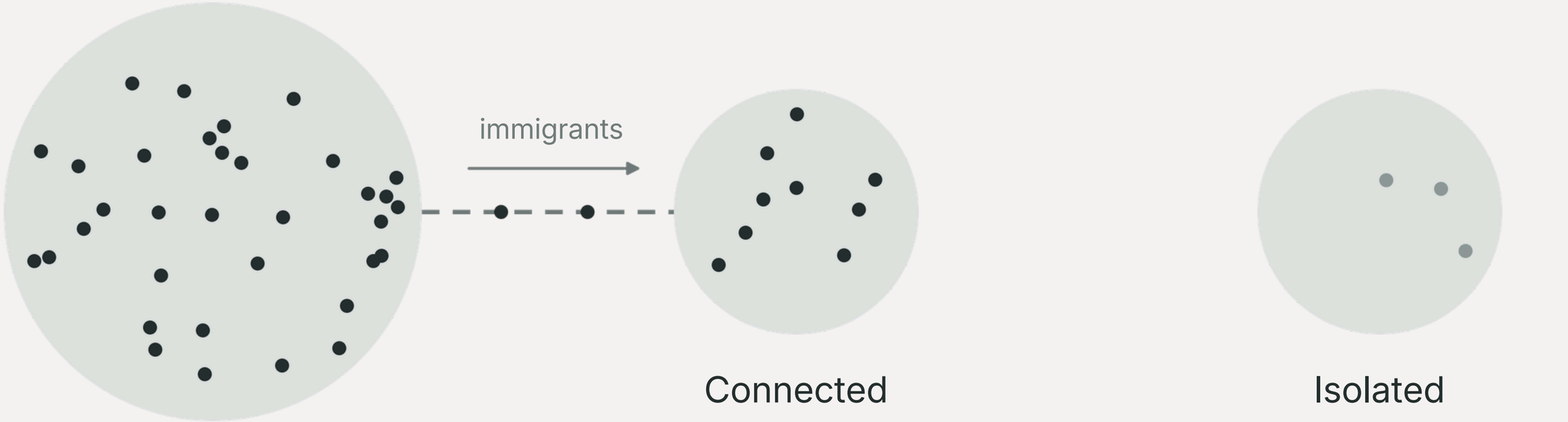
Photo: Tamás Nestor, Butterfly Conservation

## Gypsy moth

Below ≈107 males, colonies tend to shrink on their own, with chance deciding a lot

Fit: Liebhold & Bascompte 2003, Fig. 1b; regression line only. Drawn with AI assistance (Claude)

# Whether Allee effects act as a "fundamental driver" depends partly on connectivity.



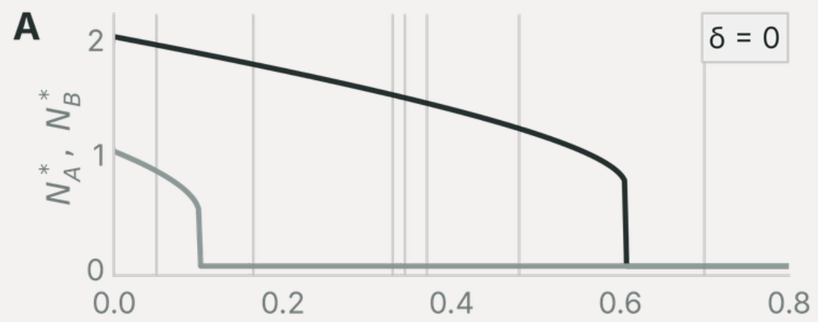

● one individual ○ a habitat patch - - dispersal

Source: Brown & Kodric-Brown 1977 · Drawn with AI assistance (Claude)

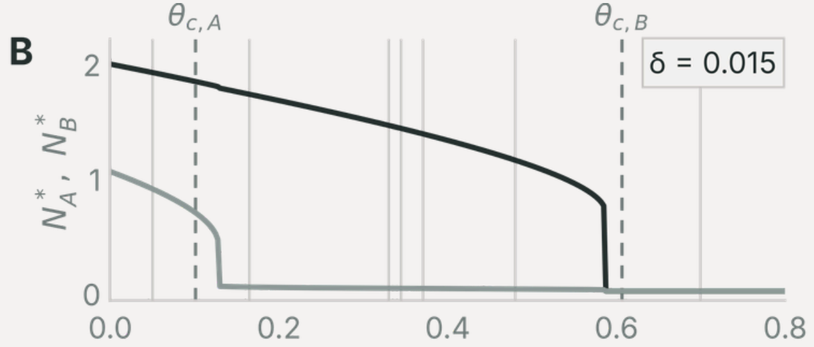
# Allee pits: pros and cons of connectivity

Population size of each patch (N*) as Allee strength θ increases

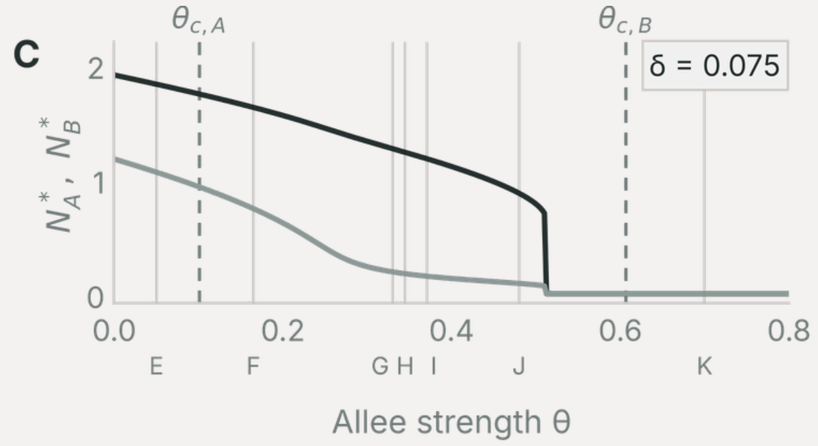
No dispersal (δ = 0)



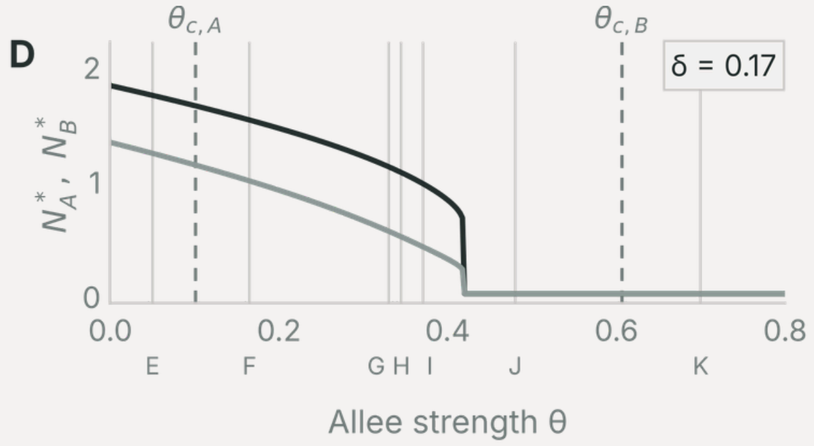

Dispersal δ = 0.015




Dispersal δ = 0.075




Dispersal δ = 0.17




Large patch B

Small patch A

θc,i: critical Allee strength (above it, an isolated patch dies out)

E–K: θ values used on the next slide

Just above θc,A the total population takes a big benefit from more connectivity and the rescue effect.

Model: Grumbach & Hilker 2025, Fig. 3A–D, recomputed. Drawn with AI assistance (Claude)

# Allee pits: reaction at different Allee strengths

Each panel: total population (up) against dispersal rate δ (across). Dashed: no dispersal.

Allee strength rises →

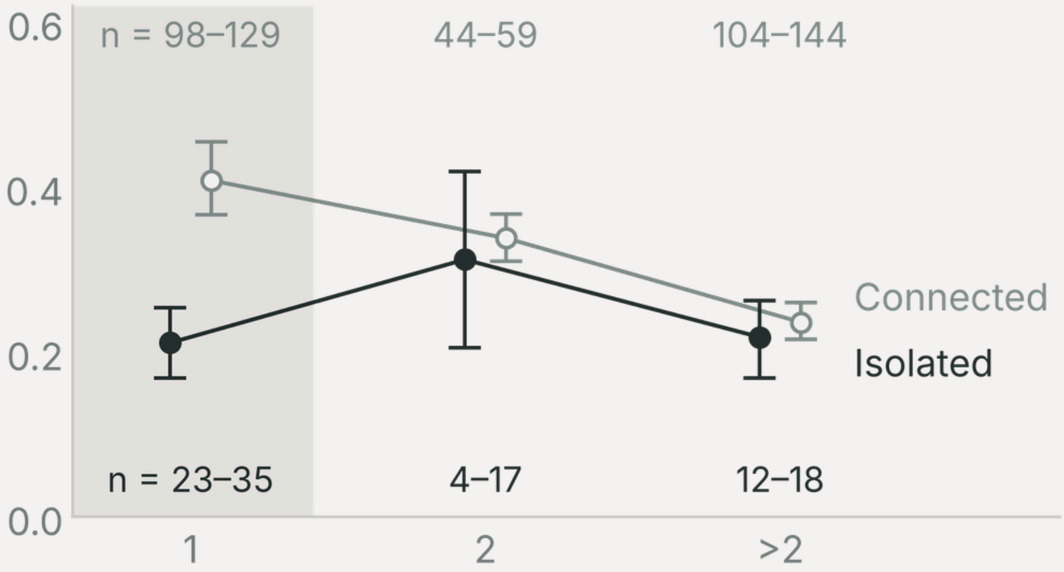
MB monotonically beneficial
UB unimodally beneficial
BTD beneficial turning detrimental
MD monotonic decrease
pit– with an Allee pit first

Allee pits occur when habitat connectivity causes a subpopulation to fall below a critical Allee threshold without getting enough rescue from neighbouring patches.

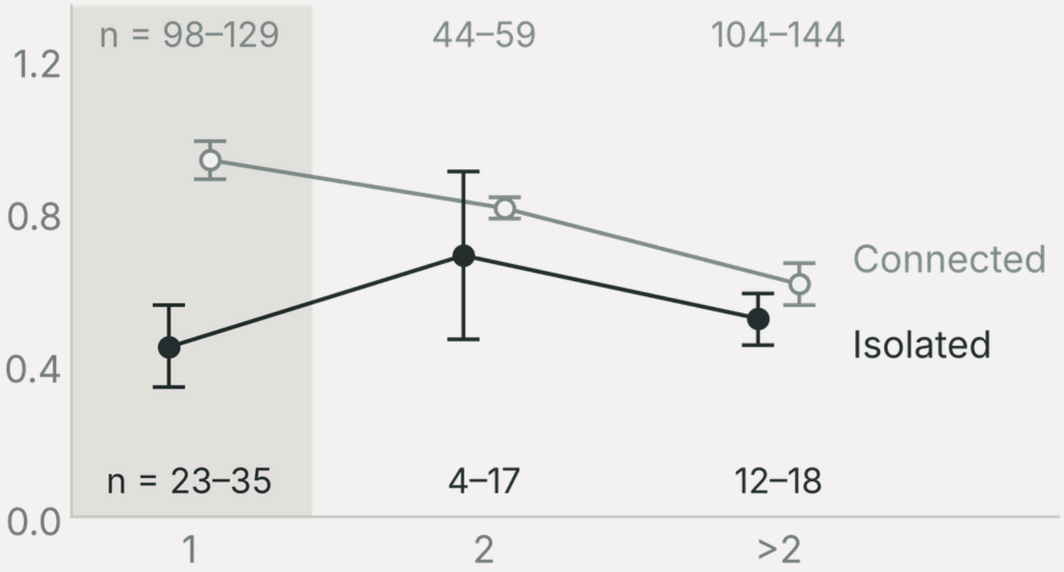
Model: Grumbach & Hilker 2025, Fig. 3E–K, recomputed. Drawn with AI assistance (Claude)

# Rescue and immigration: Glanville fritillary butterfly

## Fraction of increased populations




## Production of new larval groups per group




Mean ± 1 SE, pooled over three annual transitions (1994–96); n = range of sample sizes

One larval group: isolated populations did worse on both measures

Larger populations: no significant difference between isolated and connected

Data: Kuussaari et al. 1998, Fig. 4; values digitised, approximate. Drawn with AI assistance (Claude)

# Evidence: experiment

Setup

## Water fleas at different starting sizes, plus a predator

Result

## Small populations died out faster

Blind spot

## Shows the mechanism clearly, but only in a tank

Photo: Stephen Luk, BugGuide (© 2007)

Source: Kramer & Drake 2010

# Evidence: field data

Setup

## Fox populations on separate Channel Islands as natural replicates

Result

## Costs won: a weak Allee effect

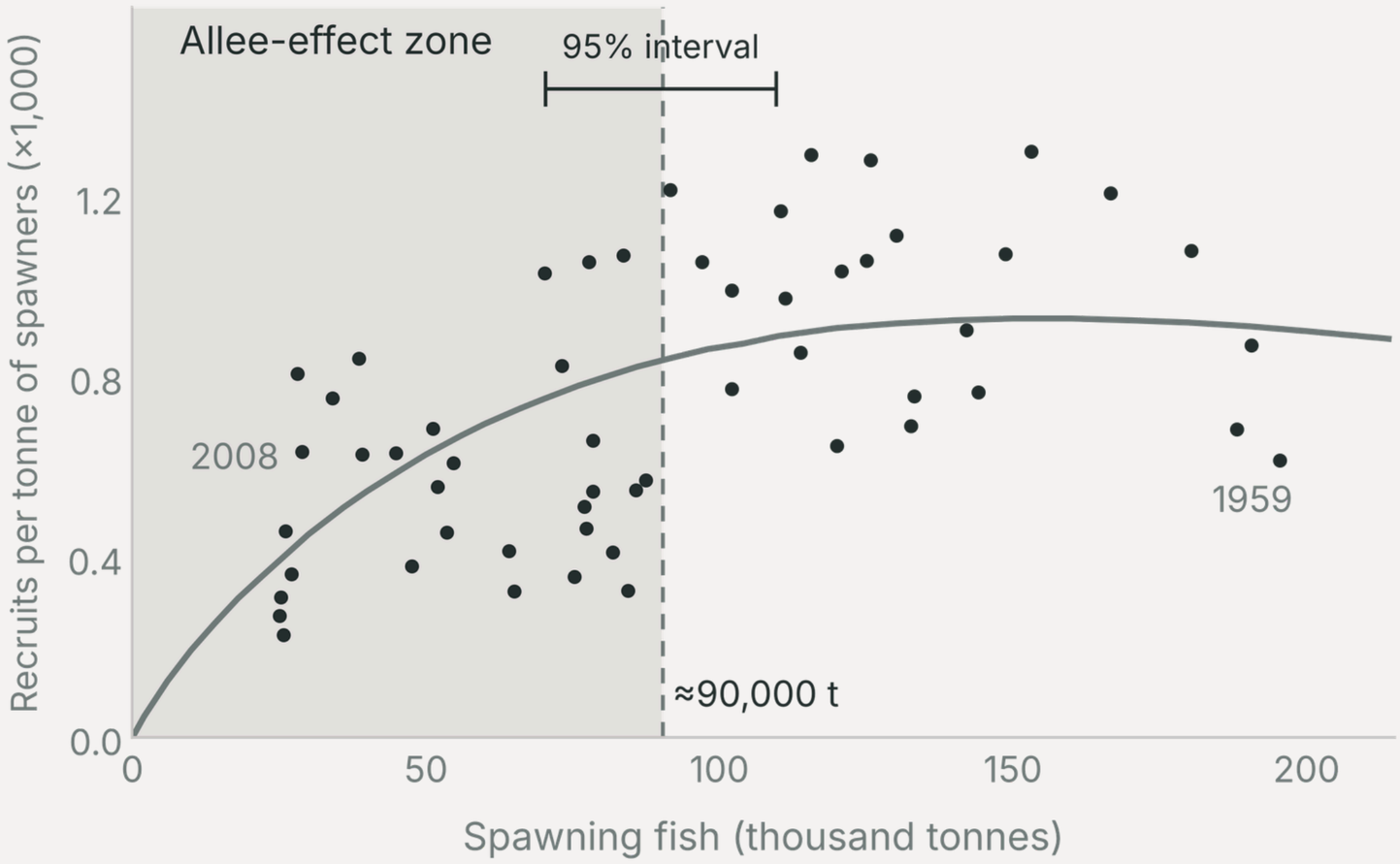
Blind spot

## Net effect depends on how the components are weighed

Photo: Maya Sugarman, KPCC / LAist

Source: Angulo et al. 2007

# Evidence: long-term records




● Each dot: one year, 1959-2008 (St Pierre Bank)

Setup

## 50 years of stock and recruitment records (1959-2008)

Not recovered after 30 years of reduced fishing

Result

## Each tonne of spawners produced fewer young

Blind spot

## The evidence only became strong after the crash

Data: Perälä et al. 2022, Fig. 2c; values digitised, approximate. Drawn with AI assistance (Claude)

Two papers, 25 years apart

# Defined in 1999, measured in 2024

## Stephens et al. 1999

Defined the terms, no new data

“Whilst theoreticians are principally concerned with demographic Allee effects, empiricists … are usually constrained to the quantification of component Allee effects.”

## Muir et al. 2024

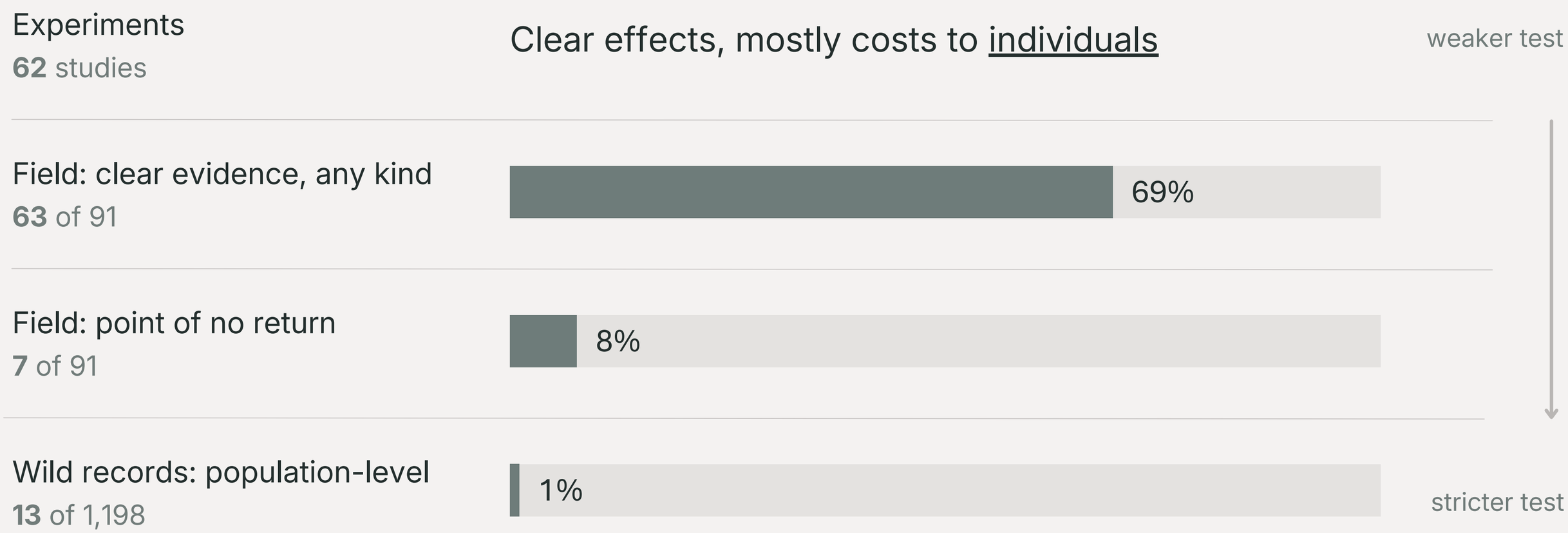
Meta-analysis of 62 experimental studies

“Much of the theoretical literature on Allee effects has focused on demographic Allee effects … In comparison, most experimental work on Allee effects has focused on component Allee effects.”

Same gap, 25 years later: we mostly measure costs to individuals, not whole populations.

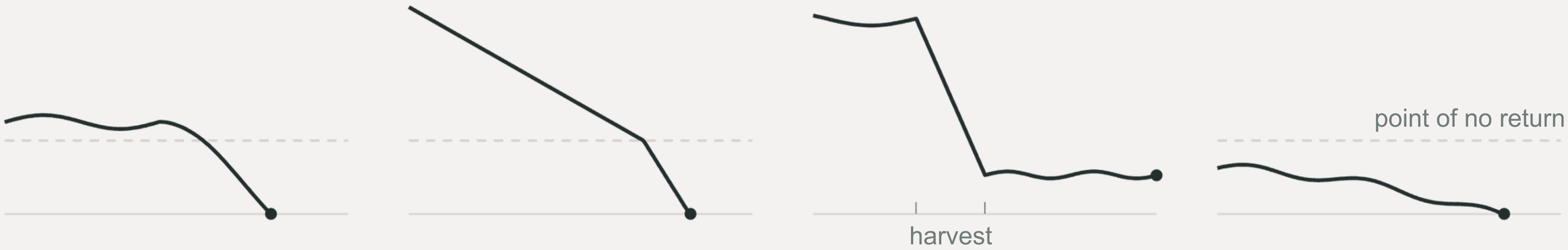
Quotes: Stephens et al. 1999, p. 187; Muir et al. 2024, p. 2

# The stricter the test, the rarer the evidence




Sources: Muir et al. 2024 · Kramer et al. 2009 · Gregory et al. 2010; separate datasets.

# Hard to see, decisive when it counts




## Very small

Chance alone can tip it under

## Declining

Slides past the threshold

## Harvested

Stays low after harvesting stops

## Newly arrived

Dies out before establishing: good news for pests

Population size over time; dashed line: point of no return (strong Allee effect)

Schematic, not data. Drawn with AI assistance (Claude)

Photo: Photaro, World History Encyclopedia (CC BY-SA 4.0)

# Neanderthals

Small population

Scattered

Possibly cut off

Photo: Photaro, World History Encyclopedia (CC BY-SA 4.0)

Cause of extinction: unproven

# Not a curiosity, but a fundamental driver.

Decides the fate of small, declining, and harvested populations, often before we can detect it.

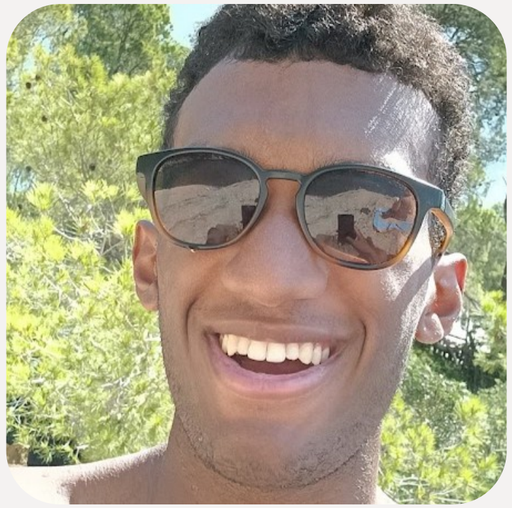
# Group

Aron Koçi

Zoe Ferencová

Laaura Mahsa Dashab

Eren Şengül



Marco Olukayode Otubushin

Cristhiam Alvarado Diaz

## Group Contributions

Aron: introduction, conclusion

Zoe: introduction, conclusion, slide design

Laaura: component, strong and weak Allee effects

Eren: demographic and strong Allee effect

Marco: rescue and immigration

Cristhiam: evidence

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● Assigned class paper

# Image credits and AI use

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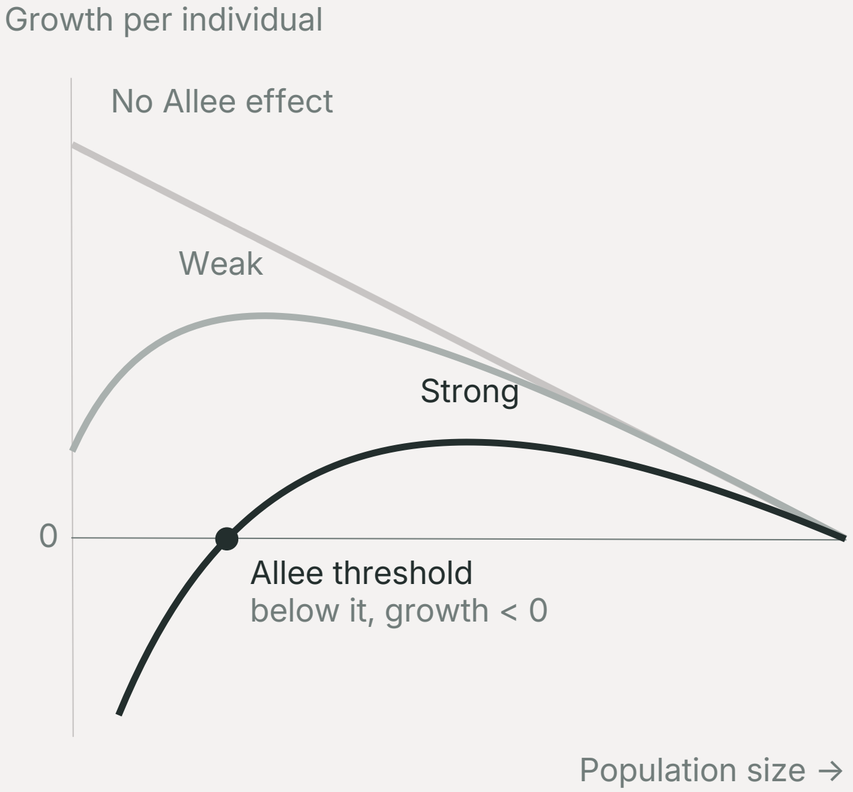
| 1, 26, 27 | Neanderthal model | Photaro, World History Encyclopedia (CC BY-SA 4.0) |
| 8 | Sea urchins | UC Irvine Samueli School of Engineering news (no credit given) |
| 8 | Vancouver Island marmots | Wired (photographer: see credit under photo) |
| 8 | African wild dogs | Conscious Explorer magazine (no credit given) |
| 8 | Bark beetles | Crested Butte News (no credit given) |
| 11 | African wild dog pups | Sabi Sabi Private Game Reserve, via Africa Endeavours |
| 13 | Flour beetle (Tribolium) | Local Hero pest control blog (no credit given) |
| 15 | Gypsy (spongy) moth | Tamás Nestor, Butterfly Conservation |
| 20 | Water fleas (Daphnia) | Stephen Luk, BugGuide (© 2007) |
| 21 | Island fox | Maya Sugarman, KPCC / LAist |
| 10 | Elephant icons | Canva |
| 2, 5, 6 | Base map of Europe | Natural Earth (public domain); Drawn with AI assistance (Claude) |

**AI use.** We used Claude to reconstruct the charts marked “Drawn with AI assistance (Claude)” from the published figures and models named in each footer. The group checked every reconstruction against the original figure. We also used AI to help us find papers, organize our sources (and keep track of them for our reference slide), and double check the facts in our final product against the papers we used.

# Study guide: key terms



| | |
|---|---|
| **Allee effect** | The smaller the population, the lower each individual's chance of surviving or reproducing. |
| **Component Allee effect** | One part of fitness, like pup survival or mating success, gets worse at low numbers.<br>Example: African wild dogs |
| **Demographic Allee effect** | The costs add up: growth per individual of the whole population falls at low numbers.<br>Example: Vancouver Island marmots |
| **Weak Allee effect** | Small populations grow more slowly, but growth stays positive. No threshold.<br>Example: Channel Island foxes |
| **Strong Allee effect** | Below a critical size, the Allee threshold, growth turns negative: more die than are born.<br>Example: Gypsy moth, ≈107 males per colony |
| **Rescue effect** | Immigrants from connected populations top up a small one. Isolated ones get no rescue.<br>Example: Glanville fritillary |




Schematic, not data. Drawn with AI assistance (Claude)

Being few also means less competition. A cost to individuals only shows up in the population if it outweighs that.

Sources: Stephens et al. 1999 · Angulo et al. 2013 · Brashares et al. 2010 · Angulo et al. 2007 · Liebhold & Bascompte 2003 · Kuussaari et al. 1998

# Study guide: how to spot each one



| Concept | Plot against population size | What you would see | Example from our talk |
|---|---|---|---|
| Component | One fitness measure (survival, mating success, pups per adult) | Rises with group or population size | Wild dogs: pup survival rose with pack size up to ≈12, with no population-level effect |
| Demographic | Growth per individual of the whole population | Lower growth at smaller sizes | Marmots: growth went negative at small sizes (38 years of counts; correlational) |
| Weak | Growth per individual at low numbers | Reduced, but still above zero | Island foxes: costs beat bigger litters, so small populations grew more slowly |
| Strong | Where growth per individual crosses zero | Negative below a threshold | Gypsy moth: growth negative below ≈107 males trapped per colony |
| Rescue | Connected vs isolated small populations | Isolated ones do worse; no gap when large | Glanville fritillary: one-larval-group populations did worse when isolated |

**Note: immigration can hurt too.** In a two-patch model, dispersal can rescue a small patch or drain one below its threshold, an “Allee pit”. Which one happens depends on how strong the Allee effect is.

Sources: Angulo et al. 2013 · Brashares et al. 2010 · Angulo et al. 2007 · Liebhold & Bascompte 2003 · Kuussaari et al. 1998 · Grumbach & Hilker 2025

# Study guide: the evidence and our answer



Three ways to look for an Allee effect

## Experiment

Water fleas + a predator: small populations died out faster

Strength Mechanism shown clearly
Blind spot Only in a tank

## Field data

Island foxes on separate islands: costs won, a weak effect

Strength Real populations
Blind spot Hard to untangle the components

## Long-term records

Atlantic cod: below ≈90,000 t, each adult produced fewer young

Strength Whole population, decades
Blind spot Evidence only strong after the crash

The stricter the test, the rarer the evidence

**62**

experiments pooled: the effect is real; its size depends on mechanism, not taxon

**7 of 91**

field studies found a point of no return (63 found some sign)

**13 of 1,198**

long-term records fit an Allee model best

Classic vs modern

Stephens 1999 defined component vs demographic. Muir 2024 measured it. Same gap: theory is about populations, experiments measure individuals.

Our answer

Not a curiosity, but a fundamental driver: it decides the fate of small, declining, harvested and newly arrived populations, often before we can detect it.

Sources: Kramer & Drake 2010 · Angulo et al. 2007 · Perälä et al. 2022 · Muir et al. 2024 · Kramer et al. 2009 · Gregory et al. 2010 · Stephens et al. 1999